

Kernel vs. User-Level Threads

Design and Performance Implications

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Fundamentals of Operating System Design (FOSD)

Learning Objectives

By the end of this topic, students should be able to:

- Explain user-level and kernel-level thread models.
- Describe how threads are created, scheduled and managed.
- Compare their advantages and limitations.
- Analyze context-switch and scheduling overhead.
- Explain blocking behavior and multicore execution.
- Evaluate which threading model is appropriate for a workload.

Why Use Threads?

Thread

A thread is an independently schedulable sequence of execution within a process.

Threads allow a process to perform multiple activities concurrently.

Benefits

- Responsiveness
- Resource sharing
- Economy compared with separate processes
- Concurrency
- Parallel execution on multicore processors

Process $\rightarrow$ {Thread 1, Thread 2, ..., Thread n}

Two Major Thread Management Models

User-Level Threads

Threads are managed primarily by a user-space thread library/runtime.

- Kernel may see one or a smaller number of execution entities.
- Thread scheduling can occur in user space.

Kernel-Level Threads

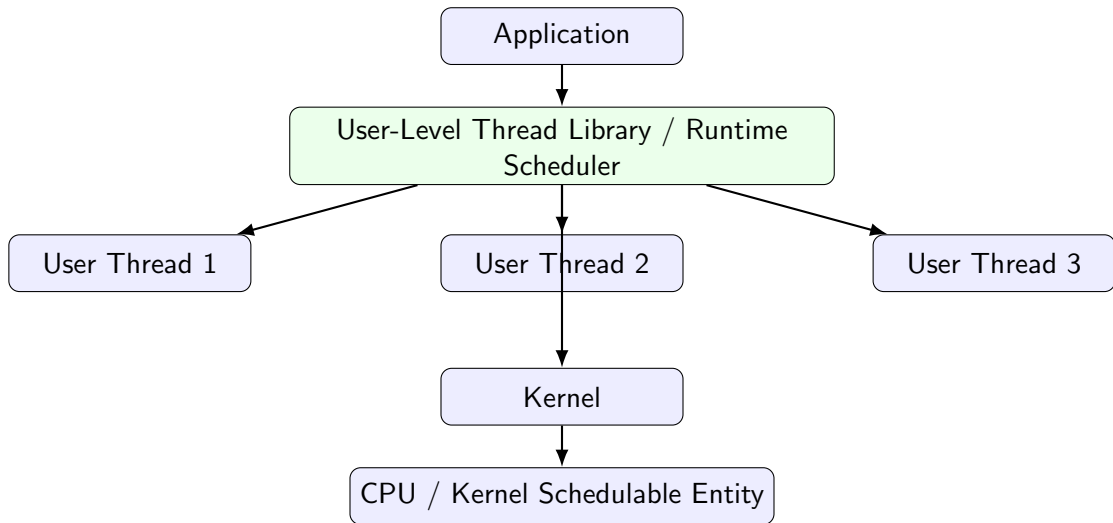
Threads are known and managed by the operating system kernel.

- Kernel schedules individual threads.
- Threads can be mapped directly to kernel schedulable entities.

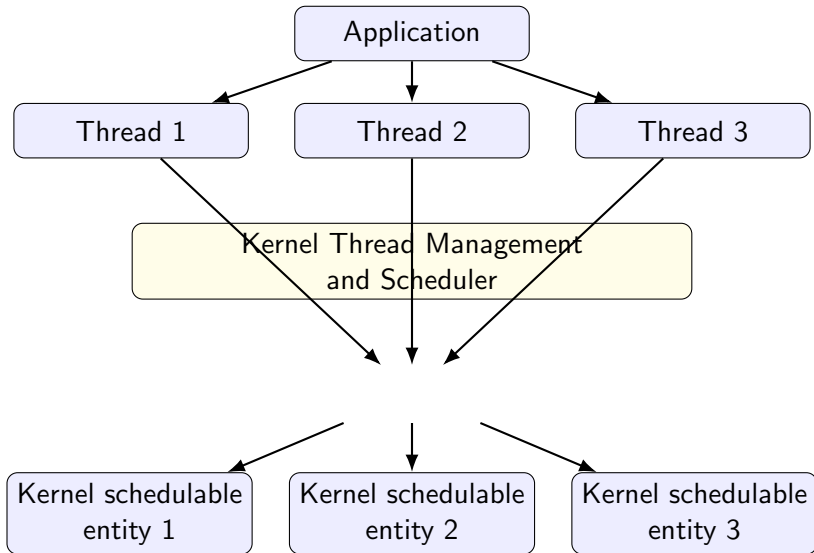
Core distinction

The key difference is **where thread management and scheduling responsibility resides**: user space, kernel space, or a combination of both.

User-Level Thread Architecture



Kernel-Level Thread Architecture



User-Level Threads: How Scheduling Works

A user-level thread library maintains information such as:

- Thread identifier
- Thread state
- Program counter
- Registers
- Stack pointer
- Thread stack
- Scheduling information

The library can implement a scheduling policy independently of the kernel.

Important

A user-space context switch can be very lightweight because it may not require entering the kernel or changing the process address space.

Kernel-Level Threads: How Scheduling Works

The kernel maintains thread-related information and schedules threads as kernel-managed entities.

Typical kernel responsibilities

- Maintain thread state.
- Select runnable threads.
- Perform preemption.
- Account for CPU usage.
- Handle blocking and wake-up.
- Schedule threads across CPU cores.

Key consequence

The kernel has direct visibility of individual threads and can schedule them independently.

User-Level vs. Kernel-Level Threads

Aspect	User-Level	Kernel-Level
Management	User library/runtime	OS kernel
Kernel awareness	Limited or indirect	Direct
Thread scheduling	User space	Kernel
Creation cost	Often low	Usually higher
Switch cost	Can be low	Usually involves kernel-managed state
Blocking system call	Can block underlying kernel entity	Other threads can continue
Multicore parallelism	Depends on mapping model	Naturally supported
Preemption	Depends on runtime/model	Kernel can preempt threads

Caution

Exact behavior depends on the threading implementation and operating system.

Major Advantage of User-Level Threads

Low management overhead

- Thread creation can avoid a kernel operation.
- User-level scheduling can be very fast.
- Context switches can be performed without entering the kernel.
- Application-specific scheduling policies can be implemented.

Example

A runtime can switch between cooperative user threads by saving and restoring their user-space execution contexts.

$$T_{\text{user switch}} \approx T_{\text{save context}} + T_{\text{select}} + T_{\text{restore context}}$$

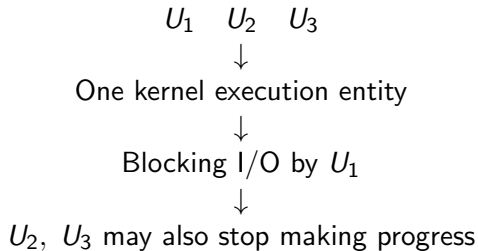
The actual cost is implementation-dependent.

Major Limitation of User-Level Threads

Blocking problem

If a user-level threading model maps many user threads onto a single kernel execution entity, a blocking system call can block that underlying entity and prevent other user threads from running.

Example



Modern runtimes can use different mappings and asynchronous I/O to mitigate this problem.

Major Advantage of Kernel-Level Threads

Kernel visibility

Because the kernel knows individual threads, it can:

- Schedule different threads on different CPU cores.
- Preempt individual threads.
- Block one thread while other threads continue.
- Account for CPU usage at thread granularity.
- Apply priorities and affinity to threads.

Multicore example

If a process contains four runnable kernel threads and the machine has four available cores, the scheduler may execute those threads concurrently, subject to OS scheduling decisions.

Kernel-Level Thread: Performance Cost

Kernel-managed threads can introduce additional overhead due to:

- Kernel scheduling operations.
- Synchronization within the kernel.
- Context switching.
- Kernel bookkeeping.
- Scheduler data-structure operations.

However, the performance comparison cannot be reduced to:

User threads = fast, Kernel threads = slow

Correct view

Overall performance depends on the workload, blocking behavior, CPU count, thread mapping, runtime design and OS implementation.

Many-to-One Thread Mapping

$$U_1, U_2, U_3, U_4 \longrightarrow K_1$$

Characteristics

- Many user threads mapped to one kernel entity.
- Very low user-level switching overhead.
- Kernel cannot independently schedule the user threads.
- A blocking operation can block the underlying kernel entity.
- No true parallel execution of those user threads on multiple cores through this single entity.

Design implication

Efficient for user-space scheduling, but limited by the single underlying kernel execution context.

One-to-One Thread Mapping

$$U_1 \rightarrow K_1 \quad U_2 \rightarrow K_2 \quad U_3 \rightarrow K_3 \quad U_4 \rightarrow K_4$$

Characteristics

- Each user-visible thread corresponds to a kernel-managed thread.
- Blocking one thread does not necessarily block other threads.
- Kernel can schedule threads independently.
- Supports genuine parallel execution on multicore systems.
- More kernel-managed resources may be required.

Many-to-Many Thread Mapping

$$U_1, U_2, U_3, U_4, U_5 \longrightarrow K_1, K_2, K_3$$

Concept

- Many user threads are multiplexed over multiple kernel execution entities.
- User runtime controls user-thread scheduling.
- Kernel schedules the available kernel entities.
- Can provide a balance between user-space flexibility and kernel-level parallelism.

Design challenge

The runtime and kernel must coordinate scheduling and blocking behavior correctly.

Thread Mapping Models: Comparison

Property	Many-to-One	One-to-One	Many-to-Many
User threads	Many	One per kernel thread	Many
Kernel entities	One	One per user thread	Several
Parallelism	Limited	Strong	Strong
Blocking isolation	Limited	Good	Good
User scheduling	Strong	Limited	Strong
Kernel overhead	Low	Higher	Intermediate/complex
Implementation complexity	Low	Moderate	High

These are conceptual models; actual OS/runtime implementations vary.

Context Switching: User vs. Kernel

User-level switch

Typically:

- 1 Save user thread registers.
- 2 Save stack pointer.
- 3 Select another user thread.
- 4 Restore its context.

May avoid a kernel transition.

Kernel-level switch

Typically involves:

- 1 Kernel scheduling.
- 2 Save current thread context.
- 3 Select another runnable thread.
- 4 Restore its context.
- 5 Return to execution.

Performance point

The cost of a switch depends on the architecture and implementation; entering the kernel is only one part of total cost.

Blocking and Concurrency

Consider three threads:

T_1, T_2, T_3

Suppose T_1 performs blocking I/O.

Single kernel entity

- T_1 blocks underlying entity.
- T_2 and T_3 may stop progressing.
- Runtime must use nonblocking/asynchronous mechanisms to avoid this.

Multiple kernel entities

- T_1 can block.
- Kernel can schedule T_2 or T_3 .
- Other threads may continue.

Design lesson

Blocking behavior is one of the most important factors when selecting a thread-management model.

Multicore Performance

For P CPU cores, effective parallel execution requires sufficient kernel-visible or otherwise appropriately mapped execution entities.

Conceptual example

With four cores:

$$T_1, T_2, T_3, T_4$$

can execute concurrently if the runtime/OS mapping exposes enough runnable execution entities to the scheduler.

Potential benefits

- Increased throughput
- Reduced completion time for parallel workloads
- Better CPU utilization

Caution

More threads do not automatically produce linear speedup because of synchronization, serial work, memory bandwidth and scheduling overhead.

Performance Evaluation Metrics

To compare threading models, measure more than thread creation time.

- **Thread creation latency**
- **Context-switch latency**
- **Scheduling overhead**
- **CPU utilization**
- **Throughput**
- **Response time**
- **Blocking/wakeup latency**
- **Scalability with increasing thread count**
- **Scalability across CPU cores**
- **Memory overhead per thread**

Simple Performance Model

Suppose a workload contains:

N thread switches

with average switching cost:

t_s

Then direct switching overhead is:

$$T_{\text{switch}} = Nt_s$$

If useful execution time is T_u :

$$T_{\text{total}} = T_u + Nt_s + T_{\text{other}}$$

where T_{other} includes synchronization, blocking, cache effects and other overheads.

$$\text{Efficiency} = \frac{T_u}{T_{\text{total}}}$$

Numerical Example: Thread Switching

Suppose:

$$N = 100,000 \text{ switches/sec}$$

User-level switch:

$$t_u = 0.5 \mu s$$

Kernel-level switch:

$$t_k = 2 \mu s$$

Direct switching overhead:

$$T_u = 100,000(0.5 \mu s) = 50 \text{ ms/s}$$

$$T_k = 100,000(2 \mu s) = 200 \text{ ms/s}$$

Thus, in this **illustrative direct-cost model**:

$$5\% \quad \text{vs.} \quad 20\%$$

Why Lower Switching Cost Does Not Always Mean Better Performance

Suppose user-level threads have lower switching overhead.

They may still perform poorly when:

- Blocking operations prevent other threads from running.
- The workload requires multicore parallelism.
- Runtime scheduling becomes a bottleneck.
- Synchronization dominates execution time.

Conversely, kernel-level threads may have greater management overhead but provide:

- Better blocking behavior.
- Kernel-controlled preemption.
- Multicore scheduling.
- Better integration with OS resource management.

Evaluation principle

Measure **end-to-end application performance**, not just thread-switch latency.

Workload-Based Design Choice

Workload	Important requirement	Design consideration
CPU-bound parallel workload	Multicore parallelism	Kernel-visible parallel execution
Interactive workload	Fast response	Efficient scheduling and low latency
I/O-heavy workload	Blocking tolerance	Multiple schedulable execution entities
Very large thread count	Low management overhead	User-space/runtime techniques may help
Real-time workload	Predictability	Scheduling guarantees and bounded latency

Conclusion

There is no universally optimal threading model; the appropriate design depends on workload and system requirements.

Design Trade-offs

- **Low overhead**

↔ Kernel visibility

- **Application-controlled scheduling**

↔ OS-controlled scheduling

- **Large number of lightweight threads**

↔ Kernel resource consumption

- **Fast user-space switching**

↔ Direct multicore scheduling and blocking support

Design principle

Threading is a systems-design problem: the best model depends on the required balance between overhead, concurrency, parallelism, responsiveness and control.

Modern Hybrid Approach

Many modern systems combine ideas from user-space runtimes and kernel scheduling.

- Application/runtime manages large numbers of logical tasks.
- Kernel provides a smaller number of schedulable execution contexts.
- Runtime maps logical work onto available execution resources.
- Asynchronous I/O can reduce blocking problems.

Many logical tasks → Runtime scheduler → Kernel execution contexts → CPU cores

Key idea

Modern concurrency systems often separate the application's logical concurrency model from the OS's physical execution resources.

Higher-Order Thinking Questions

- ① A system has thousands of user-level threads but only one kernel execution entity. What happens when one thread performs a blocking system call? Explain.
- ② A multicore server uses a thread model that provides only one kernel execution entity for many application threads. What limitation does this create?
- ③ A user-level threading system has lower context-switch overhead than a kernel-level system, but the application performs heavy blocking I/O. Explain why lower switch overhead may not produce better overall performance.
- ④ For a CPU-bound application running on 16 cores, identify the threading-model characteristics that are necessary to exploit parallelism.

Analytical Question

Consider a workload with:

1000 logical threads

running on:

8 CPU cores

Compare two designs:

- 1 All 1000 threads are mapped to one kernel execution entity.
- 2 The runtime maps the 1000 threads onto multiple kernel execution entities.

Discuss:

- Maximum possible parallelism.
- Blocking behavior.
- Scheduling overhead.
- Scalability.
- Expected application-level performance.

Challenge

Do not simply state which design is “faster.” Explain the workload assumptions required for

Summary

Concept	Key idea
User-level threads	Managed primarily by a user-space runtime/library.
Kernel-level threads	Managed and scheduled directly by the OS kernel.
Many-to-one	Many user threads mapped to one kernel execution entity.
One-to-one	Each user thread maps to a kernel-managed thread.
Many-to-many	Many user threads multiplexed over multiple kernel entities.
Performance	Depends on switching, scheduling, blocking and multicore behavior.
Multicore	Requires sufficient kernel-visible parallel execution resources.
Design choice	Must match workload and system requirements.

Key takeaway

Threading performance cannot be judged by context-switch cost alone; blocking behavior, parallelism, scheduling overhead and workload characteristics must also be considered.

Where should thread management happen?

The answer depends on the required balance among:

Low overhead + **Responsiveness** + **Parallelism**
Blocking tolerance + **Scheduling control** + **Scalability**

Choose the threading model according to the workload, not just the switching cost.